

Find the coterminal angle to $\frac{27\pi}{5}$ over the interval $[-360^\circ, 0^\circ)$.

Solution**Method 1**

$$\begin{aligned}\frac{27\pi}{5} \left(\frac{180^\circ}{\pi} \right) &= 27(36^\circ) \\ &= 972^\circ\end{aligned}$$

1 mark for conversion to degrees

$$972^\circ - (360^\circ)(3) = -108^\circ$$

$\frac{1}{2}$ mark for coterminal angle

$\frac{1}{2}$ mark for correct domain

2 marks

Method 2

$$-\frac{3\pi}{5} \text{ is a coterminal angle to } \frac{27\pi}{5}.$$

$\frac{1}{2}$ mark for coterminal angle

$\frac{1}{2}$ mark for correct domain

$$-\frac{3\pi}{5} \cdot \frac{180^\circ}{\pi} = -108^\circ$$

1 mark for conversion to degrees

2 marks

Solve the following equation over the interval $0 \leq \theta < 2\pi$.

$$(\tan \theta - 3)(\tan \theta + 1) = 0$$

Solution

$$(\tan \theta - 3)(\tan \theta + 1) = 0$$

$$\tan \theta = 3$$

$$\tan \theta = -1$$

1 mark (½ mark for each branch)

$$\theta_r = 1.249\ 046$$

$$\theta_r = \frac{\pi}{4}$$

2 marks (½ mark for each value of θ)

$$\theta = 1.249\ 046$$

$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\theta = 4.390\ 639$$

or

$$\theta = 1.249, 4.391$$

$$\theta = 2.356\ 194$$

$$\theta = 5.497\ 787$$

3 marks

$$\therefore \theta = 1.249, \frac{3\pi}{4}, 4.391, \frac{7\pi}{4}$$

or

$$\theta = 1.249, 2.356, 4.391, 5.498$$

Question 3

R10

An earthquake in Vancouver had a magnitude of 6.3 on the Richter scale. An earthquake in Japan had a magnitude of 8.9 on the Richter scale.

How many times more intense was the Japan earthquake than the Vancouver earthquake?

You may use the formula below:

$$M = \log \left(\frac{A}{A_0} \right)$$

where M is the magnitude of the earthquake on the Richter scale

A is the intensity of the earthquake

A_0 is the intensity of a standard earthquake

Express your answer as a whole number.

Solution

Method 1

Vancouver: substitute $M = 6.3$

$$6.3 = \log \left(\frac{A}{A_0} \right)$$

$$10^{6.3} = \frac{A}{A_0}$$

$$A = 10^{6.3} A_0$$

½ mark for exponential form

Japan: substitute $M = 8.9$

$$8.9 = \log \left(\frac{A}{A_0} \right)$$

$$10^{8.9} = \frac{A}{A_0}$$

$$A = 10^{8.9} A_0$$

½ mark for exponential form

To compare the two earthquakes divide their intensities.

$$\begin{aligned} \frac{\text{the intensity of Japan}}{\text{the intensity of Vancouver}} &= \frac{10^{8.9} A_0}{10^{6.3} A_0} \\ &= 398.107 \\ &= 398 \end{aligned}$$

1 mark for comparison

2 marks

Method 2

$$\frac{I_J}{I_V} = \frac{10^{8.9}}{10^{6.3}}$$
$$= 398$$

1 mark for comparison

 $\frac{1}{2}$ mark for exponential form $\frac{1}{2}$ mark for exponential form**2 marks**

Find and simplify the last term in the expansion of $(2y - 3x)^7$.

Solution

$$\begin{aligned}t_8 &= {}_7C_7 (2y)^0 (-3x)^7 \\ &= -2187x^7\end{aligned}$$

½ mark for ${}_7C_7$

½ mark for $(2y)^0$

1 mark for $(-3x)^7$ which is equal to $-2187x^7$

2 marks

Note(s):

- no deduction if ${}_7C_7$ and $(2y)^0$ are not shown

Given $\log_a 9 = 1.129$ and $\log_a 4 = 0.712$, find the value of $\log_a 12$.

Solution

Method 1

$$\log_a 9 = 1.129$$

$$\log_a 3^2 = 1.129$$

$$2 \log_a 3 = 1.129$$

$$\log_a 3 = 0.5645$$

1 mark for power rule

$$\log_a 12 = \log_a (4 \cdot 3)$$

1 mark for writing 12 as a product

$$= \log_a 4 + \log_a 3$$

1 mark for product rule

$$= 0.712 + 0.5645$$

$$= 1.2765$$

$$= 1.277$$

3 marks

Method 2

$$\log_a 12 = \log_a (\sqrt{9} \cdot 4)$$

1 mark for writing 12 as a product

$$= \frac{1}{2} \log_a 9 + \log_a 4$$

1 mark for power rule

$$= \frac{1}{2}(1.129) + 0.712$$

1 mark for product rule

$$= 1.2765$$

$$= 1.277$$

3 marks

Method 3

$$\log_a 9 = 1.129$$

$$a^{1.129} = 9$$

1 mark for exponential form

$$a = 9^{\frac{1}{1.129}}$$

$$a = 7$$

1 mark for solving for a

$$\log_7 12 = \frac{\log 12}{\log 7}$$

$$= 1.276\ 989$$

$$= 1.277$$

1 mark for value of $\log_a 12$

3 marks

Question 6

P2

How many different ways can 4 girls and 4 boys be arranged in a row if the girls and the boys must alternate?

Solution

Method 1

$$\begin{array}{cccccccc}
 & 8 & \cdot & 4 & \cdot & 3 & \cdot & 3 & \cdot & 2 & \cdot & 2 & \cdot & 1 & \cdot & 1 \\
 \nearrow & & & \nearrow & & \underbrace{\hspace{1.5cm}} & & & & & & & & & & \\
 \text{choose} & & & \text{choose} & & \text{alternate the rest} & & & & & & & & & & \\
 \text{one} & & & \text{the} & & & & & & & & & & & & \\
 \text{person} & & & \text{other} & & & & & & & & & & & & \\
 & & & \text{gender} & & & & & & & & & & & &
 \end{array}
 = 1152 \text{ ways}$$

1 mark for beginning with either gender
1 mark for alternating genders

2 marks

Method 2

$$\text{Case 1: } \frac{4}{B} \cdot \frac{4}{G} \cdot \frac{3}{B} \cdot \frac{3}{G} \cdot \frac{2}{B} \cdot \frac{2}{G} \cdot \frac{1}{B} \cdot \frac{1}{G} = 576$$

1 mark for arrangement of alternating genders

$$\text{Case 2: } \frac{4}{G} \cdot \frac{4}{B} \cdot \frac{3}{G} \cdot \frac{3}{B} \cdot \frac{2}{G} \cdot \frac{2}{B} \cdot \frac{1}{G} \cdot \frac{1}{B} = 576$$

½ mark for two cases

$$\text{Total number of ways: } 576 + 576 = 1152 \text{ ways}$$

½ mark for addition of cases

2 marks

Question 7

T3, T5, T6

Solve the following equation over the interval $[0, 2\pi]$.

$$2 \cos 2\theta - 1 = 0$$

Solution

Method 1

$$2 \cos 2\theta - 1 = 0$$

$$2(2 \cos^2 \theta - 1) - 1 = 0$$

$$4 \cos^2 \theta - 3 = 0$$

$$\cos^2 \theta = \frac{3}{4}$$

$$\cos \theta = \pm \frac{\sqrt{3}}{2}$$

$$\theta_r = \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

1 mark for identity

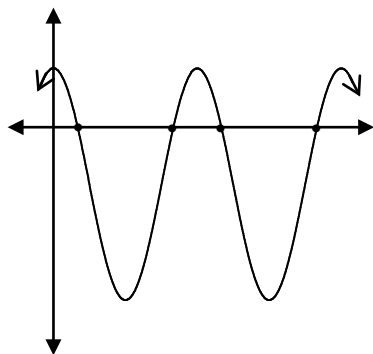
1 mark for solving for $\cos \theta$

2 marks for solutions ($\frac{1}{2}$ mark for each consistent solution)

4 marks

Method 2

$$y = 2 \cos 2\theta - 1$$



1 mark for equation

1 mark for the graph with zeroes indicated

Find all zeros over the interval $[0, 2\pi]$.

$$\theta = 0.524, 2.618, 3.665, 5.760$$

2 marks for solutions

4 marks

Solution**Method 3**

$$2 \cos 2\theta - 1 = 0$$

$$\cos 2\theta = \frac{1}{2}$$

½ mark for solving for $\cos 2\theta$

$$\text{let } x = 2\theta$$

$$\therefore \cos x = \frac{1}{2}$$

$$x_r = \frac{\pi}{3}$$

½ mark for reference angle

$$x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

1 mark for correct angles

$$\therefore 2\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

1 mark for coterminal angles

1 mark for all solutions over the interval $[0, 2\pi]$.

4 marks

Note(s):

- in Method 1, deduct a maximum of 1 mark if missing $\cos \theta = -\frac{\sqrt{3}}{2}$

Question 8

R4

Alex incorrectly explains to Rashid that the graph of $y = 2f(x) + 5$ means you first move the graph of $y = f(x)$ up 5 units and then multiply the y values by 2.

Explain to Rashid the correct way to transform the graph.

Solution

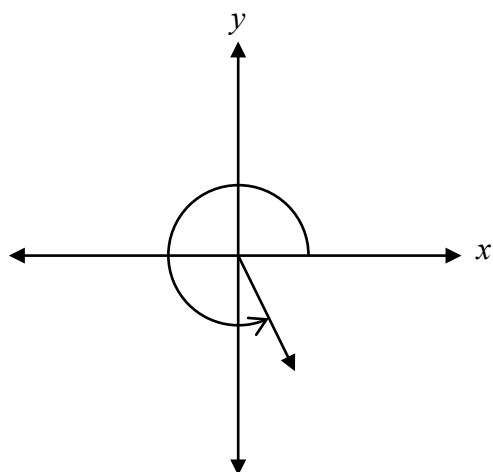
Alex explains the transformations correctly, but not in the correct order.

First multiply the y -values by 2, then move the graph up 5 units.

1 mark for explanation

1 mark

Sketch the angle of 5 radians in standard position.

Solution

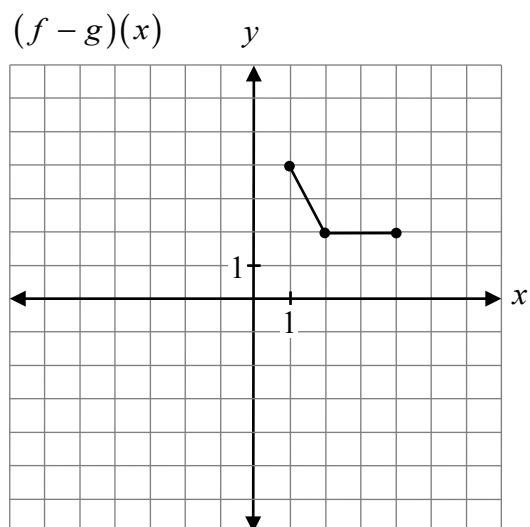
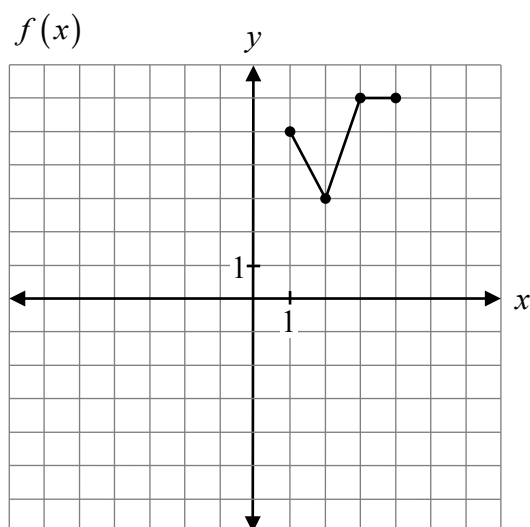
1 mark for angle drawn in Quadrant IV

1 mark

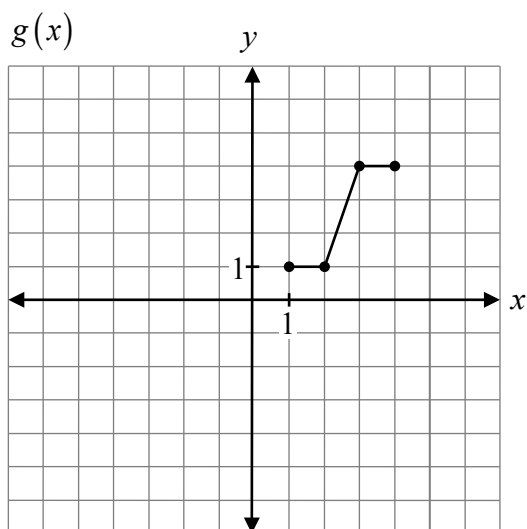
Question 10

R1

Given the graphs of $f(x)$ and $(f - g)(x)$, sketch the graph of $g(x)$.



Solution



1 mark for subtraction of $f(x) - (f - g)(x)$
 1 mark for shape representing the operation given

2 marks

A particular math class has a large number of students. From this class, you are to create a committee of 4 students that has at least 1 girl.

Without actually solving the problem, explain the strategy you would use to find the total number of ways to select this committee.

Solution

Method 1

Find all ways to form the committee and subtract the number of ways with no girls on the committee.

1 mark for identifying cases

1 mark for operation on these cases

2 marks

Method 2

Find all scenarios with girls on the committee: 1 girl on the committee, 2 girls on the committee, 3 girls on the committee, and a committee with all girls. Then add all scenarios together to get the total number of ways to create a committee with at least 1 girl.

1 mark for identifying cases

1 mark for operation on these cases

2 marks

Note(s):

- in Method 2, deduct $\frac{1}{2}$ mark if one of the cases is missing

Question 12

a) T6 b) T5, T6

a) Prove the identity below for all permissible values of θ .

$$\frac{1 + 2\cos^2 \theta}{\cos^2 \theta} = \tan^2 \theta + 3$$

b) Determine all the non-permissible values for θ .

Solution

Method 1

a)	LHS	RHS
	$\frac{1}{\cos^2 \theta} + \frac{2\cos^2 \theta}{\cos^2 \theta}$	$\tan^2 \theta + 3$
	$\sec^2 \theta + 2$	
	$\tan^2 \theta + 1 + 2$	
	$\tan^2 \theta + 3$	

$\therefore \text{LHS} = \text{RHS}$

1 mark for appropriate algebraic strategy
1 mark for appropriate identity substitutions

2 marks

b) $\cos^2 \theta = 0$
 $\cos \theta = 0$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$\therefore$ the non-permissible values of θ

are $\frac{\pi}{2} + k\pi, k \in \mathbb{I}$

or

$90^\circ + 180k, k \in \mathbb{I}$.

$\frac{1}{2}$ mark for $\cos^2 \theta = 0$

$\frac{1}{2}$ mark for any non-permissible value of θ

1 mark for all non-permissible values of θ

2 marks

Solution**Method 2**

a)	LHS	RHS
	$\frac{1 + 2\cos^2 \theta}{\cos^2 \theta}$	$\tan^2 \theta + 3$
		$\sec^2 \theta - 1 + 3$
		$\sec^2 \theta + 2$
		$\frac{1}{\cos^2 \theta} + 2$
		$\frac{1}{\cos^2 \theta} + \frac{2\cos^2 \theta}{\cos^2 \theta}$
		$\frac{1 + 2\cos^2 \theta}{\cos^2 \theta}$

1 mark for appropriate identity substitutions
1 mark for appropriate algebraic strategy

2 marks

$\therefore \text{LHS} = \text{RHS}$

b) $\cos^2 \theta = 0$

$\cos \theta = 0$

$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$

$\therefore$ the non-permissible values of θ

are $\frac{\pi}{2} + k\pi, k \in \mathbb{I}$

or

$90^\circ + 180k, k \in \mathbb{I}.$

$\frac{1}{2}$ mark for $\cos^2 \theta = 0$

$\frac{1}{2}$ mark for any non-permissible value of θ

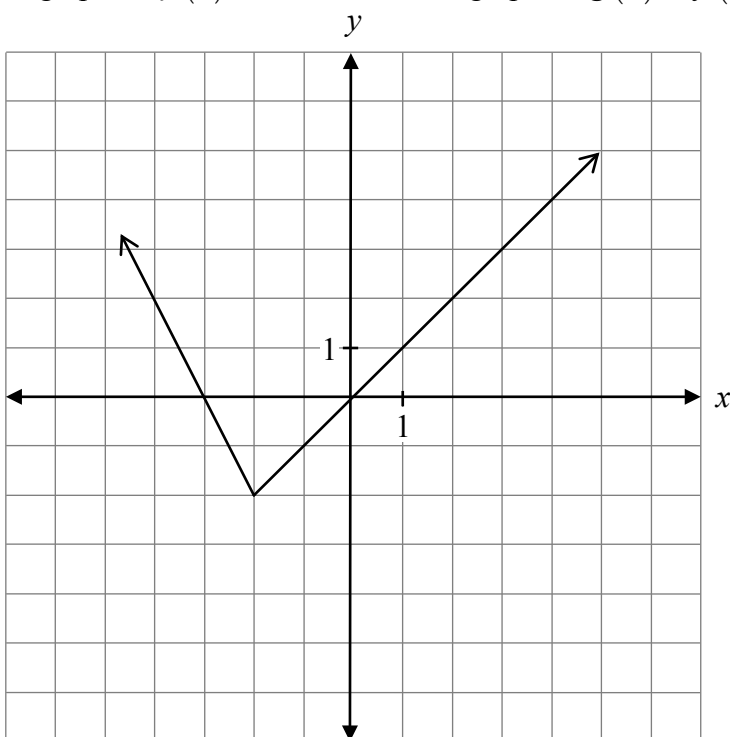
1 mark for all non-permissible values of θ

2 marks

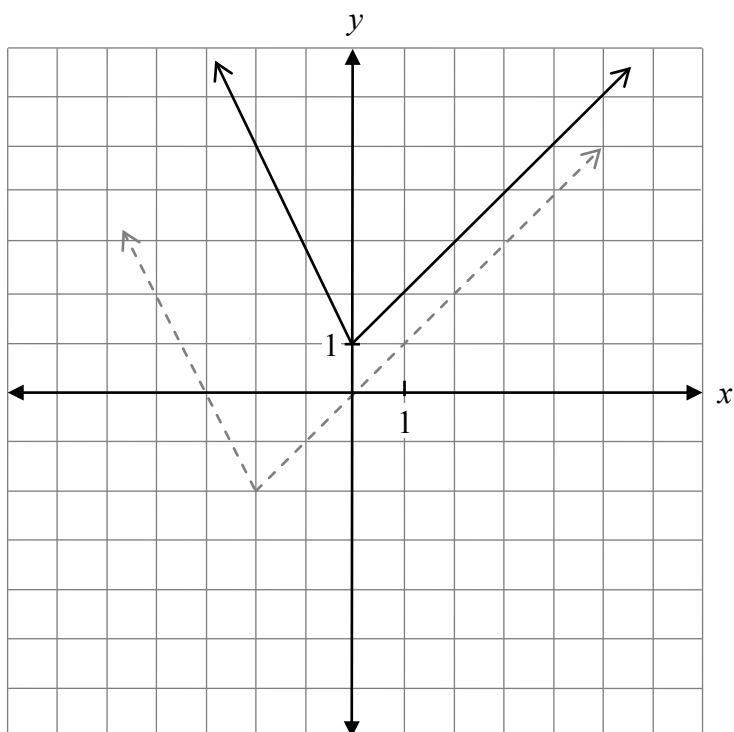
Question 13

R2

Given the graph of $f(x)$ below, sketch the graph of $g(x) = f(x - 2) + 3$.



Solution



1 mark for horizontal shift
1 mark for vertical shift

2 marks

Question 14

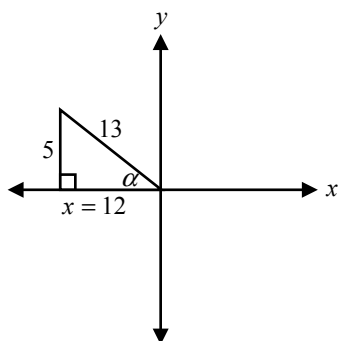
T6

Given that $\sin \alpha = \frac{5}{13}$, where α is in Quadrant II, and $\cos \beta = \frac{2}{5}$, where β is in Quadrant IV, find the exact value of:

- a) $\cos(\alpha + \beta)$ b) $\sin 2\alpha$

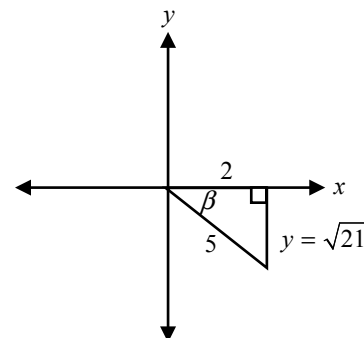
Solution

a)



$$\begin{aligned}x^2 + y^2 &= r^2 \\x^2 + 25 &= 169 \\x^2 &= 144 \\x &= \pm 12 \\x &= -12\end{aligned}$$

$$\begin{aligned}x^2 + y^2 &= r^2 \\4 + y^2 &= 25 \\y^2 &= 21 \\y &= \pm\sqrt{21} \\y &= -\sqrt{21}\end{aligned}$$



$\frac{1}{2}$ mark for value of x
 $\frac{1}{2}$ mark for value of y

$$\begin{aligned}\cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\&= \left(-\frac{12}{13}\right)\left(\frac{2}{5}\right) - \left(\frac{5}{13}\right)\left(-\frac{\sqrt{21}}{5}\right) \\&= -\frac{24}{65} + \frac{5\sqrt{21}}{65} \\&= \frac{5\sqrt{21} - 24}{65}\end{aligned}$$

$\frac{1}{2}$ mark for $\cos \alpha$
 $\frac{1}{2}$ mark for $\sin \beta$

1 mark for substitution into correct identity

3 marks

b) $\sin 2\alpha = 2 \sin \alpha \cos \alpha$

$$\begin{aligned}&= 2\left(\frac{5}{13}\right)\left(-\frac{12}{13}\right) \\&= -\frac{120}{169}\end{aligned}$$

1 mark for substitution into correct identity

1 mark

Note(s):

- accept any of the following values for x : $x = \pm 12$, $x = -12$ or $x = 12$
- accept any of the following values for y : $y = \pm\sqrt{21}$, $y = -\sqrt{21}$ or $y = \sqrt{21}$

If $f(x) = x^3$ and $g(x) = 2x - 3$, what is the value of $\left(\frac{f}{g}\right)(-1)$?

Solution

$$\begin{aligned} f(-1) &= (-1)^3 \\ &= -1 \end{aligned}$$

½ mark for substituting into $f(x)$ and $g(x)$

$$\begin{aligned} g(-1) &= 2(-1) - 3 \\ &= -5 \end{aligned}$$

$$\begin{aligned} \left(\frac{f}{g}\right)(-1) &= \frac{-1}{-5} \\ &= \frac{1}{5} \end{aligned}$$

½ mark for evaluating $\left(\frac{f}{g}\right)(-1)$

1 mark

Question 16

R3

If the point $(4, -3)$ lies on the graph of $f(x)$, which point must lie on the graph of $2f(2x)$?

a) $(8, -6)$

b) $(2, -6)$

c) $\left(8, -\frac{3}{2}\right)$

d) $\left(2, -\frac{3}{2}\right)$

Question 17

R9

The graph of $y = \log_2(2x + 6)$ intersects the graph of $y = 4$ at:

a) $x = -1$

b) $x = 1$

c) $x = 5$

d) $x = 14$

Question 18

T3

Given the point $A(-3, 5)$ on the terminal arm of an angle θ , identify the value of $\cot \theta$.

a) $-\frac{3}{5}$

b) $-\frac{5}{3}$

c) $-\frac{4}{5}$

d) $-\frac{5}{4}$

Question 19

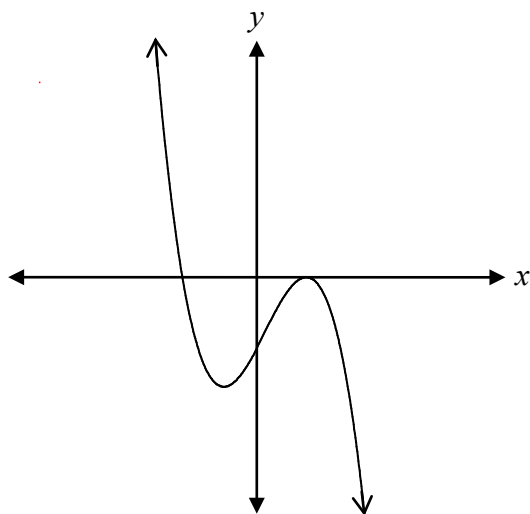
R5

The graph of $y = \left(\frac{1}{2}\right)^x$ compared to the graph of $x = \left(\frac{1}{2}\right)^y$ is a:

- a) reflection in the x -axis
- b) reflection in the y -axis
- c) reflection in the line $y = x$
- d) reciprocal function

Question 20

R12



Given the above graph of a polynomial function, which one of the following statements can be true?

- a) The function has a degree of 4 with a positive leading coefficient.
- b) The function has a degree of 4 with a negative leading coefficient.
- c) The function has a degree of 3 with a positive leading coefficient.
- d) The function has a degree of 3 with a negative leading coefficient.

Question 21

R11

Given that $(x+3)$ is a factor of polynomial $P(x)$, which of the following is true?

a) $P(-3) = 0$

b) $P(0) = -3$

c) $P(0) = 3$

d) $P(3) = 0$

Question 22

R7

Which of the following is a reasonable estimate for the value of $\log 350$?

a) 2

b) 2.5

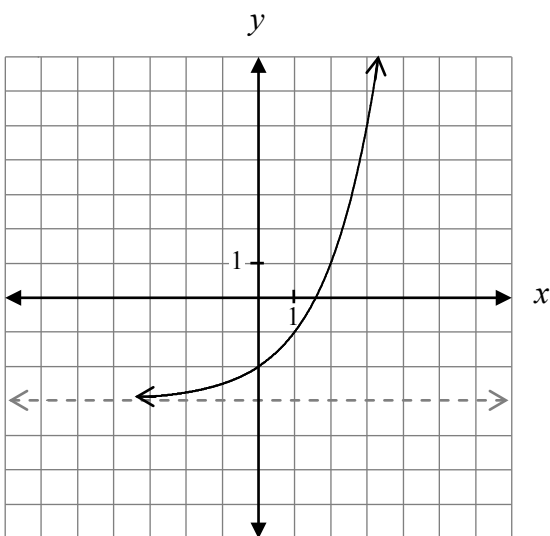
c) 2.8

d) 3

Question 23

R9

The graph of the function $f(x)$ shown below is best described by the equation:



a) $f(x) = 2^{x+3}$

b) $f(x) = 2^x + 3$

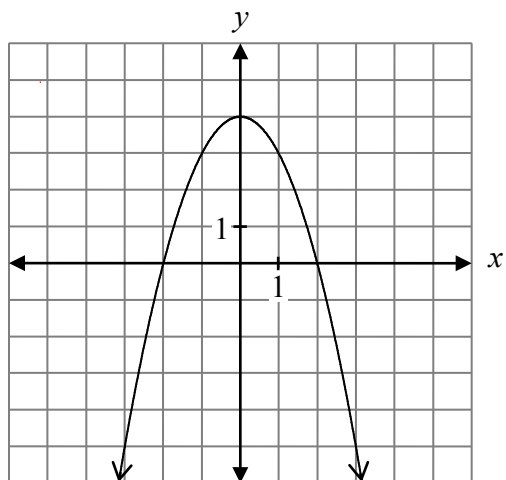
c) $f(x) = 2^{x-3}$

d) $f(x) = 2^x - 3$

Question 24

R13

Given the graph of $y = f(x)$, what is the domain of $\sqrt{f(x)}$?



a) $x \in \mathbb{R}$

b) $-2 \leq x \leq 2$

c) $x \leq -2$ or $x \geq 2$

d) $0 \leq x \leq 4$

Question 25

R10

Solve:

$$e^{\ln(5-x)} = 7$$

a) -2

b) $-\ln 2$

c) $\ln 7 - \ln 5$

d) $\frac{7}{5}$

One of the factors of $P(x) = x^3 - kx^2 - 7x + 10$ is $(x - 2)$.

Find the value of k .

Solution

Method 1

$$x = 2$$

$\frac{1}{2}$ mark for $x = 2$

$$0 = (2)^3 - k(2)^2 - 7(2) + 10$$

1 mark for remainder theorem

$$0 = 8 - 4k - 14 + 10$$

$$0 = 4 - 4k$$

$$4k = 4$$

$$k = 1$$

$\frac{1}{2}$ mark for solving for k

2 marks

Method 2

$$x = 2$$

$\frac{1}{2}$ mark for $x = 2$

2	1	$-k$	-7	10
		2	$-2k + 4$	$-4k - 6$
	1	$-k + 2$	$-2k - 3$	$-4k + 4$

$\frac{1}{2}$ mark for synthetic division

$$-4k + 4 = 0$$

$\frac{1}{2}$ mark for equating the remainder to zero

$$4k = 4$$

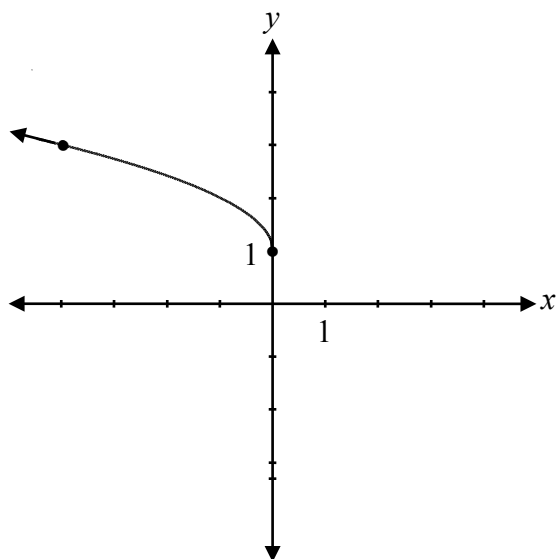
$$k = 1$$

$\frac{1}{2}$ mark for solving for k

2 marks

a) Sketch the graph of the function $y = \sqrt{-x} + 1$.

Solution

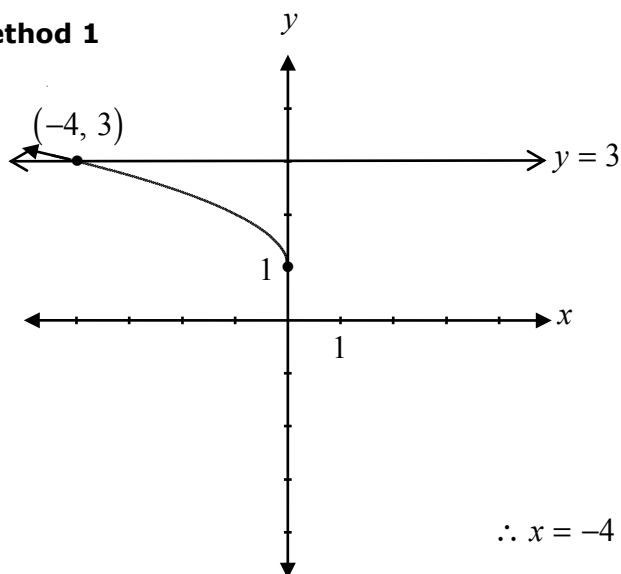


1 mark for general shape
1 mark for horizontal reflection
1 mark for vertical shift

3 marks

b) Determine the value of x when $y = 3$.

Method 1



$$\therefore x = -4$$

1 mark for consistent value of x

1 mark

Method 2

$$y = \sqrt{-x} + 1$$

$$3 = \sqrt{-x} + 1$$

$$2 = \sqrt{-x}$$

$$4 = -x$$

$$x = -4$$

1 mark for consistent value of x

1 mark

Solve the following equation:

$${}_nP_2 = {}_nC_3$$

Solution

Method 1

$$\frac{n!}{(n-2)!} = \frac{n!}{(n-3)!3!}$$

½ mark for substituting for ${}_nP_2$

$$\cancel{n!} (n-3)!3! = \cancel{n!} (n-2)!$$

½ mark for substituting for ${}_nC_3$

$$6 = \frac{(n-2)!}{(n-3)!}$$

1 mark for simplification

$$6 = \frac{(n-2) \cancel{(n-3)!}}{\cancel{(n-3)!}}$$

1 mark for expansion of $(n-2)!$

$$6 = n - 2$$

$$8 = n$$

3 marks

Method 2

$${}_nP_2 = {}_nC_3; \text{ we know } n \geq 3$$

$$\frac{n!}{(n-2)!} = \frac{n!}{(n-3)!3!}$$

½ mark for substituting for ${}_nP_2$

½ mark for substituting for ${}_nC_3$

$$\frac{n(n-1) \cancel{(n-2)!}}{\cancel{(n-2)!}} = \frac{n(n-1)(n-2) \cancel{(n-3)!}}{\cancel{(n-3)!} 6}$$

½ mark for expansion of ${}_nP_2$

½ mark for expansion of ${}_nC_3$

$$6 = \frac{\cancel{n} \cancel{(n-1)} (n-2)}{\cancel{n} \cancel{(n-1)}}$$

we can divide by n
and $(n-1)$ since
we know $n \geq 3$

$$6 = n - 2$$

$$8 = n$$

1 mark for simplification

3 marks

Solution**Method 3**

$${}_nP_2 = {}_nC_3$$

$$\frac{n!}{(n-2)!} = \frac{n!}{(n-3)!3!}$$

$$\frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}} = \frac{n(n-1)(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}3!}$$

$$6n(n-1) = n(n-1)(n-2)$$

$$6n(n-1) - n(n-1)(n-2) = 0$$

$$n(n-1)(6 - (n-2)) = 0$$

$$n(n-1)(8-n) = 0$$

$$\cancel{n-0} \quad \cancel{n-1} \quad n = 8$$

½ mark for substituting for ${}_nP_2$

½ mark for substituting for ${}_nC_3$

½ mark for expansion of ${}_nP_2$

½ mark for expansion of ${}_nC_3$

1 mark for simplification

3 marks

Note(s):

- deduct ½ mark for not rejecting non-permissible values

Given $f(x) = x^2 - 1$ and $g(x) = \sqrt{x+1}$, sketch the graph of $y = f(g(x))$ and state its domain.

Solution**Method 1**

$$f(x) = x^2 - 1$$

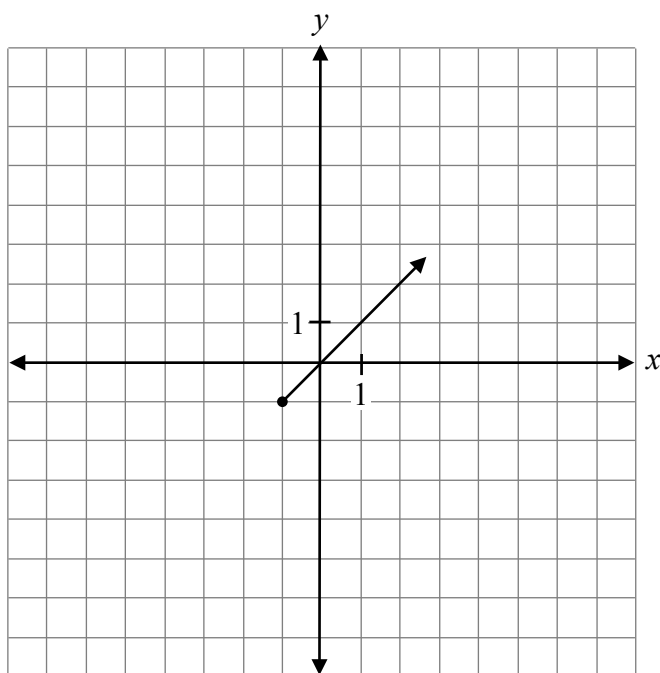
Domain: $x \in \mathbb{R}$

$$g(x) = \sqrt{x+1}$$

Domain: $[-1, \infty)$

$$\begin{aligned} f(g(x)) &= (\sqrt{x+1})^2 - 1 \\ &= x + 1 - 1 \\ &= x \end{aligned}$$

1 mark for determining the function $f(g(x))$



1 mark for graph of composite function

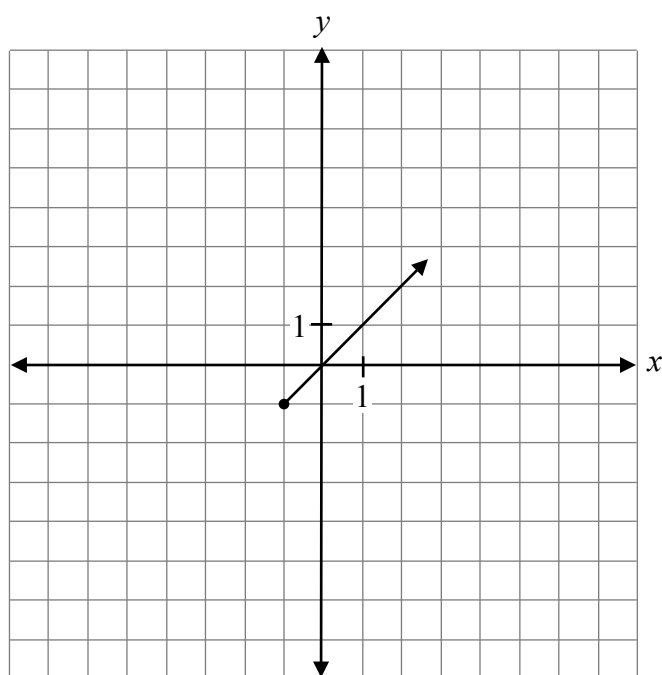
Domain of $f(g(x)) : [-1, \infty)$ **or** $\{x | x \geq -1, x \in \mathbb{R}\}$ 1 mark for restricted domain

3 marks

Solution**Method 2**

x	$g(x)$	$f(g(x))$
-2		
-1	0	-1
0	1	0
1		1
2		2
3	2	3

1 mark for table of values



1 mark for graph of composite function

Domain of $f(g(x)) : [-1, \infty)$ or $\{x | x \geq -1, x \in \mathbb{R}\}$

1 mark for restricted domain

3 marks

Write the equation of the horizontal asymptote for the function $f(x) = \frac{x-3}{x-2}$.

Solution

$$y = 1$$

1 mark for equation of horizontal asymptote

1 mark

The x -intercept of $f(x)$ is 4 and the x -intercept of $g(x)$ is 4.

Benjamin concludes that the x -intercept of $f(x) + g(x)$ is 8.

Do you agree with Benjamin? Justify your answer.

Solution

No, I do not agree with Benjamin.

Benjamin added the x -values instead of adding the y -values.

If the x -intercept of $f(x)$ is 4, then $y = 0$.

If the x -intercept of $g(x)$ is 4, then $y = 0$.

$\therefore$ the x -intercept of $f(x) + g(x)$ is 4.

1 mark for justification

1 mark

Solve the following equation:

$$2 \log_4 x - \log_4 (x + 3) = 1$$

Solution

$$2 \log_4 x - \log_4 (x + 3) = 1$$

$$\log_4 \left(\frac{x^2}{x+3} \right) = 1$$

$$4^1 = \left(\frac{x^2}{x+3} \right)$$

$$4(x+3) = x^2$$

$$x^2 - 4x - 12 = 0$$

$$(x-6)(x+2) = 0$$

$$x = 6 \quad \cancel{x = -2}$$

1 mark for power rule

1 mark for quotient rule

1 mark for exponential form

$\frac{1}{2}$ mark for solving for x

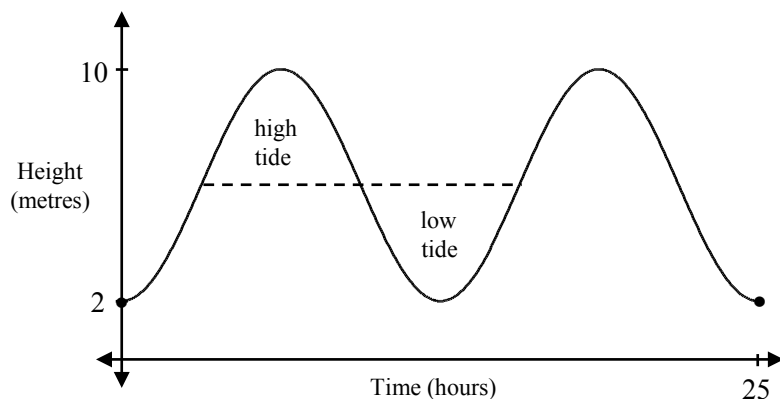
$\frac{1}{2}$ mark for rejecting extraneous root

4 marks

Question 33

T4

The following graph represents tidal levels in the Bay of Fundy over a 25-hour period.



- What is the average height of the water?
- What is the period of the graph above?

Explain what the period represents in this situation.

Solution

- 6 metres

1 mark

$$\begin{aligned} \text{b) Period} &= \frac{25}{2} \\ &= 12.5 \text{ hours} \end{aligned}$$

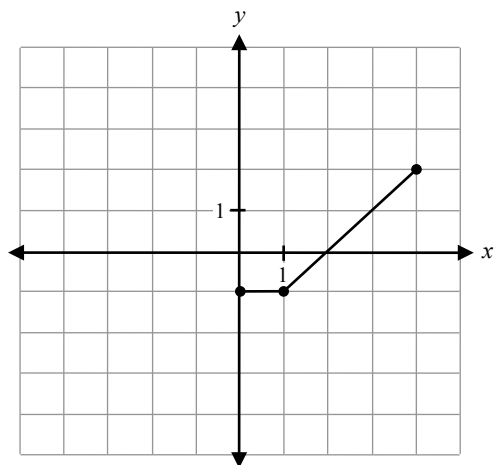
1 mark for period

The period represents the time to complete one cycle of tidal levels in the Bay of Fundy.

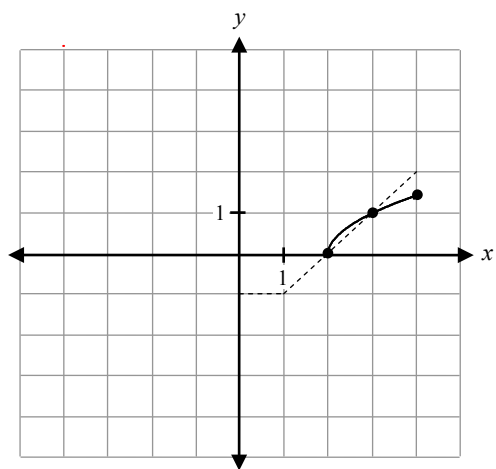
1 mark for explanation

2 marks

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



Solution



1 mark for restricting the domain

$\frac{1}{2}$ mark for graph above $y = f(x)$ over the range $[0, 1]$

$\frac{1}{2}$ mark for graph below $y = f(x)$ over the range $[1, 2]$

2 marks

When $P(x)$ is divided by $x - 3$, it has a quotient of $2x^2 + x - 6$ and a remainder of 4.

Determine $P(x)$.

Solution

$$P(x) = (x - 3)(2x^2 + x - 6) + 4$$

1 mark for polynomial $P(x)$

or

1 mark

$$P(x) = 2x^3 - 5x^2 - 9x + 22$$

Identify the domain and range of the following function:

$$f(x) = \frac{3}{x^2 + 1}$$

Solution

Domain: $\{x \in \mathbb{R}\}$

1 mark for domain

or

$(-\infty, \infty)$

Range: $\{y \in \mathbb{R} \mid 0 < y \leq 3\}$

1 mark for range

or

$(0, 3]$

2 marks

Evaluate:

$$\csc\left(\frac{11\pi}{6}\right) + \sin^2\left(-\frac{3\pi}{4}\right) + \cos\left(\frac{23\pi}{3}\right)$$

Solution

$$= (-2) + \left(-\frac{\sqrt{2}}{2}\right)^2 + \frac{1}{2}$$

$$= -2 + \frac{1}{2} + \frac{1}{2}$$

$$= -1$$

1 mark for $\csc\left(\frac{11\pi}{6}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

1 mark for $\sin^2\left(-\frac{3\pi}{4}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

1 mark for $\cos\left(\frac{23\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

3 marks

Evaluate the coefficient of the term containing x^3 in the expansion of $(1+x)^7$.

Justify your answer.

Solution

Method 1

				1				
			1		1			
		1		2		1		
	1		3		3		1	
	1	4		6		4		1
1		5	10		10	5		1
1	6		15	20		15	6	1
1	7	21	<u>35</u>	35	21	7	1	

1 mark for justification

The coefficient of x^3 is 35.

1 mark for identifying coefficient

2 marks

Method 2

$$t_4 = {}_7C_3 (1)^4 (x)^3$$

1 mark for justification

$${}_7C_3 = \frac{7!}{3!4!}$$

$$= \frac{7 \cdot \cancel{6} \cdot 5 \cdot \cancel{4}!}{\cancel{3}! \cancel{4}!}$$

$$= 35$$

1 mark for evaluating coefficient

2 marks

The coefficient of x^3 is 35.

Which of the following equations could be solved without the use of logarithms?

Without actually solving the problem, explain your choice.

$$4^x = 10^{3x+1} \quad \text{or} \quad \left(\frac{1}{3}\right)^{2x+1} = 27^{4x-1}$$

Solution

$\left(\frac{1}{3}\right)^{2x+1} = 27^{4x-7}$ can be solved without the use of logarithms because $\frac{1}{3}$ and 27 can both be changed to a base of 3.

1 mark for explanation

1 mark

Sketch the graph of $y = x^3 + x^2 - 5x + 3$ given that one of the x -intercepts is 1.

Identify the x -intercepts and y -intercept.

Solution

$$x = 1$$

1	1	-5	3
1	2	-3	0

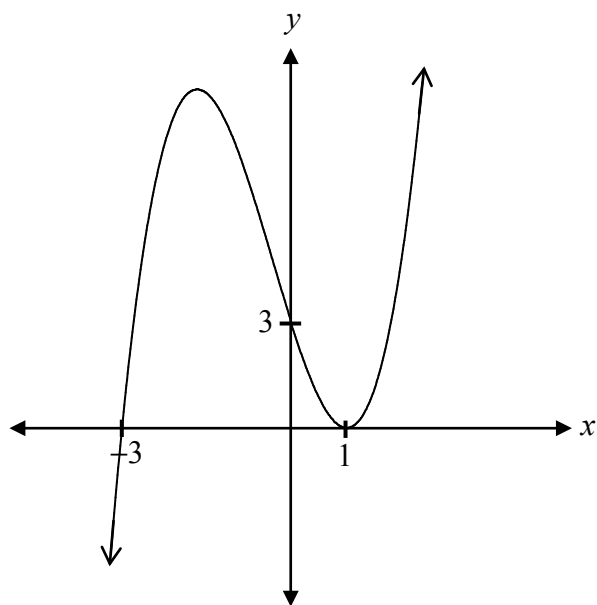
1 mark for synthetic division

$$y = (x - 1)(x^2 + 2x - 3)$$

$$y = (x - 1)(x + 3)(x - 1)$$

$$y = (x + 3)(x - 1)^2$$

1 mark for identifying the factors



2 marks for graph ($\frac{1}{2}$ mark for x -intercepts, $\frac{1}{2}$ mark for multiplicity, $\frac{1}{2}$ mark for y -intercept, $\frac{1}{2}$ mark for end behaviour)

4 marks

If $f(x) = \frac{1}{x-2}$ and $g(x) = x-2$, what is the domain of $f(x) \cdot g(x)$?

Solution

$$f(x) = \frac{1}{x-2} \qquad g(x) = x-2$$

Domain: $x \in \mathbb{R}, x \neq 2$ Domain: $x \in \mathbb{R}$

Domain of $f(x) \cdot g(x)$: $\{x \in \mathbb{R} \mid x \neq 2\}$

1 mark for domain of $f(x) \cdot g(x)$

1 mark

Given $f(x) = (x+1)^2$ for $x \leq -1$, write the equation of $y = f^{-1}(x)$.

Solution

Method 1

$$y = (x+1)^2$$

$$x = (y+1)^2$$

$$y = \pm\sqrt{x} - 1$$

1 mark for inverse

½ mark for solving for y

Since the domain of $f(x)$ is $x \leq -1$,
the range of the inverse is $y \leq -1$.

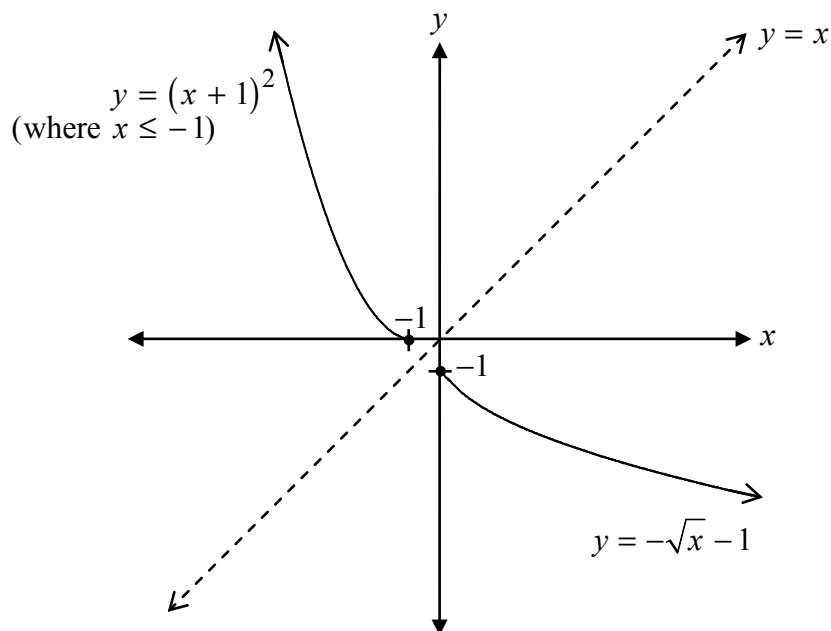
$$\therefore y = -\sqrt{x} - 1$$

$$f^{-1}(x) = -\sqrt{x} - 1$$

½ mark for rejecting $y = \sqrt{x}$

2 marks

Method 2



1 mark for reflection over the line $y = x$

1 mark for correct equation

2 marks

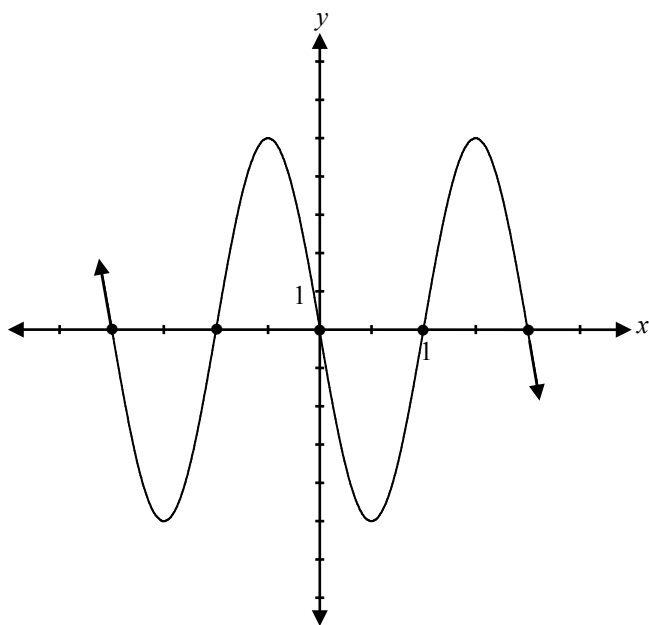
Sketch a graph of at least one period of the function $y = 5 \sin[\pi(x+1)]$.

Clearly indicate the x -intercepts.

Solution

$$b = \pi$$

$$\therefore \text{period} = \frac{2\pi}{\pi} = 2$$



1 mark for amplitude
1 mark for horizontal shift
1 mark for period
1 mark for clearly indicating at least two x -intercepts consistent with graph

4 marks

Sketch the graph of the following function:

$$f(x) = \frac{x-2}{(2x-3)(x-2)}$$

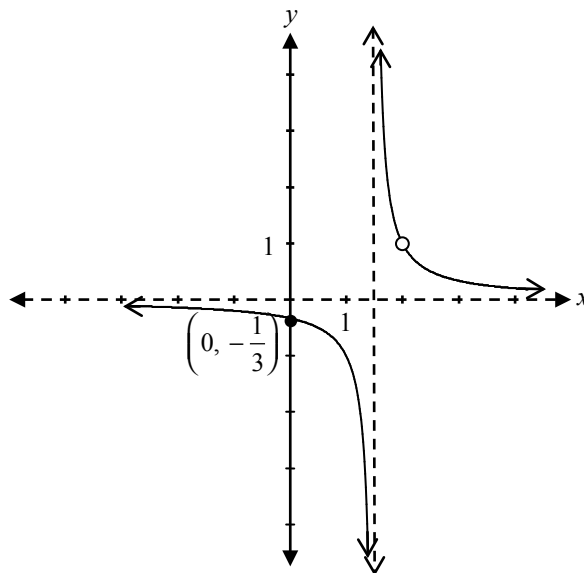
Solution

$$\begin{aligned} f(x) &= \frac{x-2}{(2x-3)(x-2)} \\ &= \frac{1}{2x-3} \text{ with a point of discontinuity at } x=2 \end{aligned}$$

point of discontinuity: $f(2) = 1$

$\therefore$ there is a point of discontinuity at $(2, 1)$

$$\begin{aligned} \text{y-intercept: } f(0) &= \frac{0-2}{(2(0)-3)(0-2)} \\ &= -\frac{2}{6} \\ &= -\frac{1}{3} \end{aligned}$$



1 mark for horizontal asymptote at $y = 0$

1 mark for vertical asymptote at $x = \frac{3}{2}$

$\frac{1}{2}$ mark for graph left of vertical asymptote

$\frac{1}{2}$ mark for graph right of vertical asymptote

1 mark for point of discontinuity at $(2, 1)$; ($\frac{1}{2}$ mark for $x = 2$, $\frac{1}{2}$ mark for $y = 1$)

4 marks